

AI-Generated Steganographic Malware: Evaluating Detection and Evasion in Modern Anti-Virus Systems

Master of Science in Computer Science at the University of
Johannesburg

Student: Thendo Siphuma

Supervisor: Prof Daniel Du Plessis

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Chapter 2: Literature review

2.1. Introduction

Recent advances in generative Artificial Intelligence (AI) (for example, transformer and diffusion models) have not only improved learning experience in education field (Morales-Chan, et al. 2024) but has enabled the expansion of the attack surface in cybersecurity enabling digital concealments such as AI-Generated Steganography as noted by Varney (2019). Generative AI provides cyber adversaries with minimal programming skills a platform to automate the creation of complex malicious code, facilitate malware generation, and conduct sophisticated cyberattacks with reduced technical effort (Metta, et al. 2024). Recent studies by Motwani, et al. (2024) demonstrate how cyber adversaries can leverage AI to automate data hiding, reduce technical skills required to create strong and undetectable threats.

Steganography derived from the Greek words *steganos* (covered) and *graphia* (writing) refers to the art of hiding information (Kour and Verma 2014) in carriers such as media files in such a way that the presence of the information is unknown (Mandal, et al. 2022). Steganography forms part of a larger part of data hiding techniques which also include watermarking (Rustad et al., 2023). Within Steganography three primary categories are commonly identified - text, audio, and image steganography (Singh 2016). This study will focus on image steganography due to its extensive use in hiding malicious payloads while keeping its visual quality and structural integrity (Saputro, et al. 2024).

Steganography techniques directly undermine the effectiveness of traditional antivirus (AV) systems as shown by Dell'Orco, et al. (2025b) on their test on antivirus against elusive malware. Conventional AVs primarily rely on signature-based detection where at least one byte of the code will be compared to an existing signature of already existing malwares, which are stored in a database known (Chakravarty, et al. 2019). Although some of the modern tools incorporate heuristics, rules and AI analysis their performance remains questionable when confronting malwares concealed in benign-looking files (Raheel Abbas and Hussain 2025). This gap provides a basis for the need for advanced hybrid mechanisms capable of identifying hidden or AI-generated malware.

This chapter reviews literature from 2000 to 2025 from peer-reviewed journals, industry white papers, and public datasets. It traces the evolution of steganography and its intersection with artificial intelligence, examines the limitations of conventional antivirus solutions, and analyses the detection challenges posed by stegomalware. Finally, it explores emerging hybrid detection frameworks that combine YARA rule-based analysis with machine learning (ML) techniques to improve the detection of AI-generated steganographic malware.

2.2. Evolution of Steganography

Steganography is the science that involves communicating secret data in an appropriate multimedia carrier such as an image (Cheddad, et al. 2010). The practice has a rich history and literature which has rapidly developed over the past decades (Sahu and Sahu 2020). Although its roots go back to ancient times when Demaratus (king of Sparta) and Histiaeus (ruler of Miletus) communicated using hidden messages (Sahu and Sahu 2020), the recent rise in technological advancements have moved the practice from inks to digital pixel manipulations (Evsutin, et al. 2020). Understanding this evolution is crucial as it highlights how technological change has continuously increased the detection challenges of steganography for modern detection systems (Bamanga and Babando 2024). Understanding this evolution is crucial, as each technological phase has introduced new concealment techniques that continuously challenge modern detection systems (Kumar, et al. 2024).

The evolution of Steganography can be looked at through three interrelated perspectives that shows how the discipline has advanced alongside computing technologies. These include carrier modality, capacity–detectability trade-offs, and the degree of automation in embedding which collectively reveal how steganography has evolved from simple, human-crafted concealment methods to complex, intelligent, and data-driven systems capable of deceiving modern forensic detection tools as noted below:

Carrier modality:

This dimension explains the evolution of how information has been carried through steganography over time from ancient methods, such as hiding messages under wax tablets as described by (Sahu and Sahu 2020), to modern techniques that use digital media like images, audio, and video fuelled by the rise of digital technologies. Different advanced frameworks now further allow multiple media simultaneously without extensive neural network training, thereby reducing computational costs and improving adaptability (Song, et al. 2024). This expanded range of carrier modalities has increased the potential attack surface, making detection increasingly complex for cybersecurity systems (Mallick and Nath 2024).

Capacity-detectability trade-offs:

A fundamental challenge throughout the history of steganography is achieving an optimal balance between data-hiding capacity and noticeability (Sahu and Sahu 2020). Increasing the amount of data hidden within a carrier file generally introduces more statistical distortions, making the presence of the hidden message easier to detect (Cheddad, et al. 2010). Early digital methods, such as simple Least Significant Bit (LSB) substitution, could achieve high capacity but were vulnerable to statistical steganalysis attacks that could easily spot these modifications (Sahu and Sahu 2020). Frequency-domain methods such as Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT) improved robustness against image processing and

detection, though at the cost of reduced capacity of data hidden (Cheddad, et al. 2010). Modern AI-based models, particularly those employing Generative Adversarial Networks (GANs), address this trade-off by learning embedding patterns that mimic the natural statistical features of carrier images, thereby increasing capacity while minimizing detectability (Liu, et al. 2020). The transition from rule-based to data-driven embedding signifies a significant enhancement in steganographic strength (Luo, et al. 2024).

Degree of automation in embedding:

Early digital methods required substantial human intervention (Sahu and Sahu 2020) but in recent steganography techniques, it has transitioned from manual operations to fully automated, intelligent systems (Majeed, et al. 2021). The beginning of adaptive steganography introduced partial automation by allowing algorithms to analyse cover images and select regions often textured or noisy areas where data could be concealed with minimal impact (Cheddad, et al. 2010). With the rise of deep learning, complete automation has become possible with models such as Generative Adversarial Networks (GANs) and Convolutional Neural Networks (CNNs) autonomously learning from optimal embedding strategies without predefined rules (Saputro, et al. 2024). For example, a GAN-based system trains a generator network to create stego-images realistic enough to deceive a discriminator, thereby producing highly imperceptible and secure embeddings (Liu, et al. 2020). This high level of automation significantly enhances the stealth and complexity of steganographic malware, posing major challenges for traditional antivirus detection systems (Dell'Orco, et al. 2025b).

2.2.1. Historical and Traditional Techniques

In today's world, steganography continues to play a crucial role in ensuring secure and private communications, particularly in environments subject to surveillance and monitoring (Choudhary and Agarwal 2024). This role has actually advanced from ancient times as Sahu and Sahu (2020) explained Demaratus wrote the warning message to Greece about the forthcoming attack inside the piece of wood, which was placed beneath a wax tablet. Likewise, Histiaeus tattooed a message on his messenger's shaved scalp and then waited for the hair to regrow (Diepenbroek 2023) and In World War II invisible ink and microdots were popular methods of hiding information within letters and photographs (Hammad, et al. 2024).

Digital steganography emerged with the ability to manipulate carrier bits commonly using images and audio files to embed secret payloads without obvious visible change (Kumar, et al. 2024). Digital image data is a prevalent medium for steganography due to its high redundancy, providing ample capacity for embedding images (Abdalwahid, et al. 2024). This study focusses on Image steganography, which can be categorized into two domains:

- Transform domain which hides information by modifying the frequency components of a media file, such as using discrete cosine or wavelet transforms, making it more robust against compression and attacks than spatial domain methods (Alyousuf, et al. 2020) and
- Spatial domain steganography (Mandal, et al. 2022), In spatial domain techniques the secret message is hidden inside the image by applying some manipulation over the different pixels of the image steganography (Swain and Lenka 2014) as shown in the figure below:

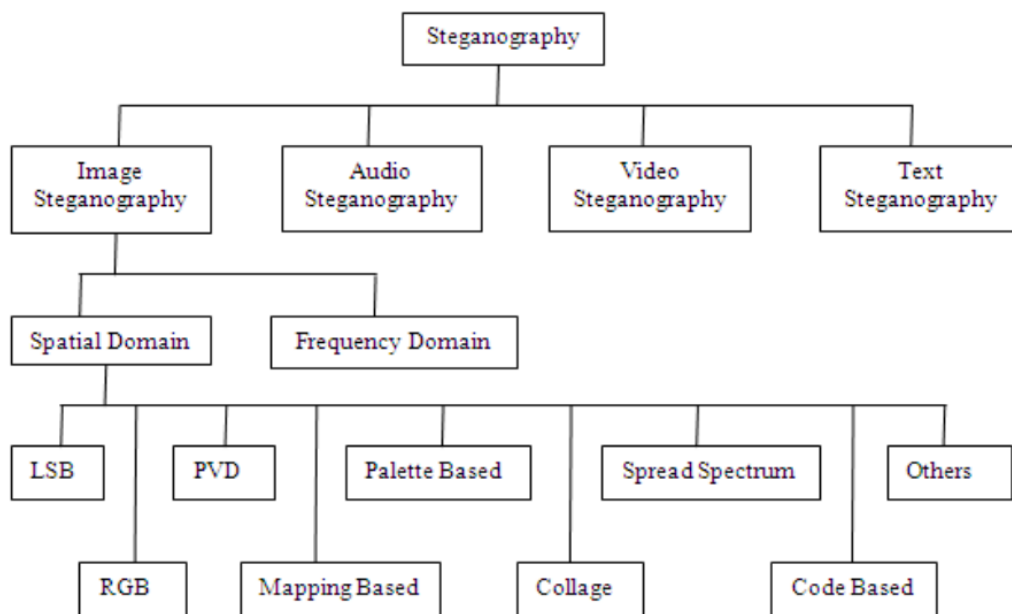


Figure 1: (Swain and Lenka 2014) Classification of spatial domain image steganography techniques

Swain and Lenka (2014) categorize spatial-domain steganography into several techniques, including the, including Least Significant Bit (LSB) method, Pixel Value Differencing (PVD), and RGB-based embedding:

- Least Significant Bit (LSB) : Described by Swain and Lenka (2014), LSB is the simplest and popular image steganographic method is the least significant bit (LSB) substitution and It embeds messages into cover image by replacing the least significant bits directly. To be further under 2.2.2. Least Significant Bit.
- Pixel value differencing: Described by Mandal, et al. (2022) conceals secret data by leveraging the fact that edge regions of an image can tolerate greater modification than smooth areas. In this approach, the cover image is divided into non-overlapping pixel pairs, and the secret message is embedded in the pixel-value difference, calculated as $B_i = k_{i+1} - k_i$ where B_i ranges from 0 to 255.

However, this study excludes PVD because it allows for lower data capacity compared to LSB, which distributes information uniformly across all pixels.

- Other spatial-domain techniques, such as RGB-Based hides data in the red, green, and blue channels of pixels using modified LSB approaches to enhance security and minimize visual distortion (Swain and Lenka 2014).

Alongside embedding techniques, steganalysis focuses on detecting, analysing, and potentially extracting hidden information from media to uncover the presence of steganography (Eid, et al. 2022). Steganalysis is often framed using warden models, which define how adversaries interact with hidden communication (Al-Obaidi, et al. 2024). The active warden model proactively disrupts steganography by modifying potential cover objects such as images before they can be used, thereby causing embedding algorithms to fail or produce easily detectable artefacts (Desoky 2011) and passive warden model merely monitors communication channels or files to detect hidden content without altering the transmitted data (Luo, et al. 2024). This study aligns with the passive warden model, as it analyses files non-destructively by examining the bit patterns of the least significant bits (LSBs) across digital images (Aslam, et al. 2022).

2.2.2. Least Significant Bit (LSB) Substitution

The simplest and most popular image steganographic method is the least significant bit (LSB) substitution (Emad, et al. 2018) which embeds messages into cover images by replacing the least significant bits directly (Swain and Lenka 2014). Although this study focuses on LSB embedding in images, the same principle can be applied to other media types such as audio and video, which are also effective steganographic carriers due to their inherent redundancy (Asad, et al. 2011). The simplicity and high embedding capacity of LSB-based approaches make them foundational to modern spatial-domain steganography (Alhomoud 2021).

To reduce detectability and enhance security, several parts of LSB have been developed. One such approach is such as randomized LSB, which modifies pixel insertion positions to avoid predictable sequential patterns (Rafat and Sajjad 2024). ALabaichi, et al. (2020) described how randomization is a common strategy that improves resistance to detection by distributing hidden data irregularly across the image and in relation to this, Rahman, et al. (2025) propose a Magic Matrix approach that shuffles blocks of an image's blue channel before embedding, enhancing resistance to steganalysis. Other methods employ adaptive selection, which targets specific regions of an image, like edges detected via the Canny algorithm, to hide information (Aslam, et al. 2022). Adaptive methods can also vary the number of bits hidden in each pixel to improve the stego-image's quality (Emad, et al. 2018). LSB matching (LSBM) is another widely known variant that alters the embedding logic to be less statistically detectable (ALabaichi, et al. 2020).

Despite its simplicity, Rahman, et al. (2022) showed that the LSB method remains highly vulnerable to statistical steganalysis especially when detection techniques such as histogram analysis and RS (Regular-Singular) testing are introduced as they can expose the subtle distributional shifts introduced by LSB embedding. A high degree of similarity between the histograms indicates minimal distortion, while significant differences can reveal tampering (ALabaichi, et al. 2020). A more powerful and targeted technique is RS steganalysis, a statistical method specifically used to identify the presence of hidden data in LSB-based steganography (Rahman, et al. 2022). Numerous studies confirm that standard LSB substitution methods are susceptible to RS detection and are therefore considered among the easiest steganographic techniques to compromise these limitations highlight the importance of developing adaptive and intelligent variants that can resist statistical and AI-based detection (ALabaichi, et al. 2020).

2.2.3. AI-Generated Steganography

AI-generated steganography uses neural encoders, generative models, or optimization loops to conceal payloads that are difficult to detect inside digital media (Raymond and Rick 2023). This evolution represents a subtle yet potent cyber threat where malicious code is concealed within seemingly innocuous files, rendering traditional cybersecurity defences often powerless (Ali and Bhatti 2024). A notable approach, known as coverless steganography, directly transforms secret data into a photorealistic image rather than embedding it within a pre-existing cover file (Rehman, et al. 2024). Generative models such as Generative Adversarial Networks (GANs) and diffusion models are central to this process, using hidden information to construct the final stego media (Liu, et al. 2020). Within a GAN framework, the competition between the generator and discriminator acts as a loop that continually refines the carrier's realism and stealth, making AI-generated steganographic malware highly resistant to conventional detection (Saputro, et al. 2024).

Fig. 1

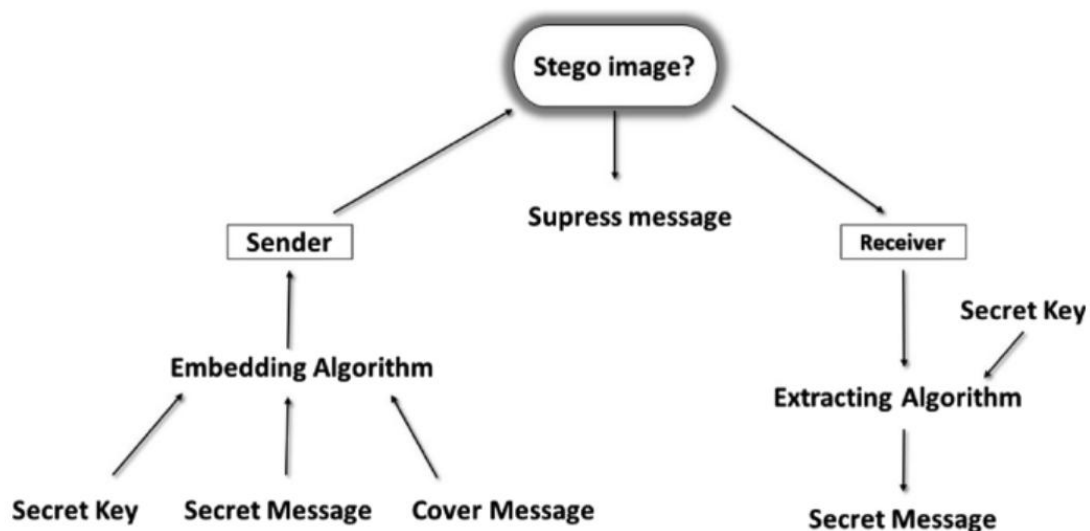


Figure 2 : (Georges and Magdi 2020) Architecture of the steganography model

Encoder-decoder architectures learn mappings between payloads and carrier disquiets, while GANs can synthesize carriers that look natural and embed information during the generating process (Zhang, et al. 2021). In Encoder-decoder models, an encoder can be explicitly trained to embed a secret payload into an image, while a corresponding decoder is trained to extract it, creating a structured method for concealment (Gao, et al. 2022). In contrast, GAN-based concealment produces a stego-image that is visually very natural and difficult to distinguish from an authentic one (Saputro, et al. 2024). This is a significant departure from traditional techniques like Least Significant Bit (LSB) substitution, which operate at the bit or pixel level by modifying an existing image. AI-powered methods instead embed data at a semantic or structural level, making the concealment an integral part of the media's fabric (Raymond and Rick 2023). Because this strategy leaves no conventional artefacts for signature-based tools like YARA to detect, it poses a challenge to modern antivirus systems (Almomani, et al. 2022b).

AI enables semantic-level embeddings that preserve contextual coherence and can be optimized against detection models, thereby increasing stealth and cross-detector transferability (Raymond and Rick 2023). Operating at this semantic level allows malicious data to be integrated within the meaning of the content itself without disrupting its logical or structural consistency. Chang and Echizen (2025) demonstrate how information can be hidden in the linguistic domain of audiovisual files, preserving semantic equivalence and rendering the concealment resistant to spatial or temporal analysis. Such semantic plausibility makes it difficult for existing machine-learning-based anomaly detectors to recognize statistical irregularities. Moreover, adversarial training further strengthens model robustness against steganalysis tools explained by Gao, et al. (2022) who further note As a result, detection accuracy for some AI-generated stego-images drops to approximately 50 percent which is equivalent to randomly guessing which highlight the urgent need for more advanced detection frameworks.

Despite their power, AI-driven methods lack standardized benchmarks and raise reproducibility questions, complicating assessments of Anti-Virus resilience (Dell'Orco, et al. 2025b). The primary forensic challenge in detecting AI-generated steganographic malware is that there is often no "original" carrier file for comparison, which undermines traditional detection methods that rely on identifying statistical deviations (Raymond and Rick 2023). This complicates both YARA rule creation and the training of anomaly detection models, which benefit from a clear baseline of what constitutes a "clean" file. This problem is magnified when threat actors use private or fine-tuned AI systems, as it becomes nearly impossible to reverse-engineer the logic to create reliable detection signatures (Raymond and Rick 2023). As a result, AV systems often fail to detect malware concealed through steganographic techniques (Almomani, et al. 2022b). The ability for these techniques to carry viruses and malware presents major risks, justifying the need for a hybrid detection pipeline that combines

signature-based and anomaly-based approaches to counter this evolving threat (Williams 2019).

2.3. Limitations of Current Antivirus Solutions

As demonstrated in the previous sections, AI-generated steganography integrates malicious data within benign files producing realistic and undetectable files (Williams 2019). These advances challenge traditional antivirus (AV) systems which primarily rely on static signature-based or heuristic scanning (Raheel Abbas and Hussain 2025) as concealment leaves no recognizable code fragments or metadata anomalies as shown by Dell'Orco, et al. (2025b). This section critically evaluates these limitations, focusing on how conventional detection architectures fail against AI-enhanced steganographic malware and why hybrid, adaptive mechanisms are now required.

2.3.1. Ineffectiveness of Signature-Based Detection

Signature based detection works by looking for certain known byte sequences, code fragments, or digital fingerprints in a large database of signatures which is one of the key process in any virus system (Al-Asli and Ghaleb 2019). These techniques struggle to work when confronted with AI-generated steganography which embeds payloads in ways that leave no detectable signatures (Raymond and Rick 2023). Modern generative models such as Generative Adversarial Networks (GANs) and encoder–decoder architectures produce stego-images that appear entirely authentic by concealing data within the semantic or structural composition of media file as noted by (Saputro, et al. 2024) and (Gao, et al. 2022). Unlike traditional Least Significant Bit substitution methods that alter existing pixel data (Rahman, et al. 2022), these AI-driven methods transform entire images leaving no byte patterns or modification artefacts that signature-based systems can recognize (Raymond and Rick 2023).

It is possible to preserve semantic equivalence between the cover and stego transcripts reducing the likelihood of detection as malicious content is seamlessly integrated within the context of the host file (Chang and Echizen 2025). This creates a serious problem during detection, since there's usually no original version of the file to compare against and without that baseline, it becomes nearly impossible to verify authenticity using traditional statistical or signature-based methods (Raymond and Rick 2023). As a result, antivirus programs that rely on fixed signatures or rule-based detection often can't spot these hidden payloads. Research shows that when malicious code is hidden inside multimedia files, traditional antivirus tools frequently miss them sometimes failing to detect as much as half of the concealed threats (Almomani, et al. 2022b).

2.3.2. Failure Against Concealed Malware

While Williams (2019) highlights that the detection of AI-generated steganography is becoming increasingly difficult for cybersecurity experts, exposing organizations to

significant risks, Joyce, et al. (2023) provide the technical basis with their findings showing challenges with some deep malware analysis methods as they are computationally expensive and cannot feasibly scale to the volume of files processed by antivirus engines. In one study, while many antivirus engines detected a ransomware sample in its original format, none could identify it after it was concealed within an HEVC video stream (Almomani, et al. 2022b). Similarly, image-based concealment has been shown to bypass detection by 90% against commercial antivirus tools (Dell’Orco, et al. 2025b). This evasion is effective because deep scanning is prone to high rates of false positives, as many legitimate applications perform image manipulation operations that are structurally similar to steganographic decoding as further shown by Dell’Orco, et al. (2025b). Attackers take advantage of this weakness by hiding harmful code inside ordinary-looking files, such as JPEG images, allowing them to slip past security software undetected (Knöchel and Karius 2024) and as a result, steganography tools and the malware hidden within them can easily slip past perimeter defences such as web filters and antivirus systems (Frith 2007).

A primary advantage of stegomalware is its ability to bypass static analysis because the malicious payload is only revealed at runtime (Dell’Orco, et al. 2025b). This creates a critical time-to-detection problem, as experiments show it can take several days for AV engines to adapt their signatures after a new threat is submitted Dell’Orco, et al. (2025b) further explained. This operational window allows the malware to establish a connection to its Command & Control (C&C) servers to download additional malicious code (Knöchel and Karius 2024) and attackers often take advantage of this delay to quietly steal sensitive information, using steganographic methods to move files or confidential company data out of the organization without being noticed (Frith 2007). Because the malicious activity only commences after the payload is extracted from its benign-looking carrier, behaviour-based detection systems are placed at a disadvantage, reacting to the attack only after the initial infiltration has succeeded (Frith 2007).

2.4. Detection Challenges for Steganographic Malware

Steganography can be used to secretly share the data for legitimate purpose such as digital watermarking, privacy-preserving communication, and copyright protection (Bamanga and Babando 2024), but adversaries increasingly misuse these same techniques to conceal malicious payloads and evade detection mechanisms (Chaganti, et al. 2021). Identifying these kind of stegomalware is especially difficult for security systems because the hidden data blends in well that it looks just like the harmless content of the original file (Alazab and Pham 2023).

Data can be hidden in multiple levels, from imperceptible pixel-level modifications in images to small semantic manipulations in AI-generated text, audio, and video, making

it hard for conventional antivirus and steganalysis tools to detect (Luo, et al. 2024). The wide variety of file types, encodings, and encryption methods further expands the detection search space and increases both computational cost and false-positive rates (Sanjalawe, et al. 2025). Thus, detecting steganographic malware demands approaches that can analyse large datasets and integrate statistical, behavioural, and intelligent anomaly-detection techniques (Wang, et al. 2023). This section looks at the main technical and practical challenges that make it difficult to reliably detect this growing steganography threat.

2.4.1. Ineffectiveness of Existing Tools

There has been increasing use of security software such as antivirus (AV) software which has improved the general resistance in systems (Knöchel and Karius 2024). However, most commercial antivirus solutions primarily rely on signature-based detection, identifying malware only after its signature code or binary patterns has been analysed and catalogued (Dell'Orco, et al. 2025b). Since steganography principle is to hide secret information without being noticed by a third party by modifying redundant regions in digital media, this mechanism is inherently ineffective against the practice and as a result, steganographic content can pass undetected through conventional signature-based defences (Wu, et al. 2021).

Suarez-Tangil, et al. (2014) demonstrated in their study that the current static malware detection techniques fail to identify malicious components when they are embedded separately from the code for example, when the payload is concealed within an image. Their findings showed that the malicious part only becomes active when the file is run, so even if the asset is scanned beforehand, no static signature can detect it. Similarly, Badhani and Muttoo (2018) evaluated ten commercial anti-malware programs to see if they could detect malicious applications hidden inside images using various techniques, nine of the ten programs were unable to detect any of the hiding techniques, while the most effective one still failed against malware hidden with encryption or steganography.

A more recent study by Dell'Orco, et al. (2025a) confirms that existing AV systems remain ineffective against stegomalware. Although the proposed solution combines steganalysis with dynamic analysis to improve detection of stegomalware, developing a detection mechanism that operates independently of specific embedding schemes remains an open challenge.

2.4.2. Evasion Techniques

Morden stegomalware incorporates different adversarial evasion strategies to bypass machine learning based and steganalysis models and Ghamizi, et al. (2021) identified several approaches that demonstrate how attackers adapt embedding strategies to defeat evolving detection systems:

EAST (Evasion Attack Steganography):

Regarded as a new steganography and watermarking technique, it works by encoding secret messages within adversarial images generated through a multi-label targeted evasion attack ensuring that the resulting stego-images are visually convincing and optimised to mislead steganalysis and image-classification models simultaneously.

ADV-EMB:

This technique combines the traditional statistical heuristics embedding (such as UNIWARD) against a detector. The technique embeds the payload using statistical methods, then iteratively modifies the image to reduce detector confidence, making it less detectable in the feature.

SGAN (Steganographic Generative Adversarial Networks):

It adds the steganalyser to the conventional GAN framework that will see the difference between a clean and stego image. The generator is then trained to deceive both the steganalyser and the discriminator producing cover images that are more realistic and evasive.

Hidden:

This technique introduces differentiable noise layer between the encoder and decoder, allowing the model to learn embeddings to common image transformations such as compression, resizing, or noise. This allows the generation of stego-images that remain undetectable even after real-world distortions.

These evasion techniques show how deep learning allows attackers to create flexible and advanced image concealment strategies that can encode and decode information with high precision and minimal distortion (Sadeq and Kapure 2023). The growing challenge of these techniques underscores the need for an approach that combines rule-based and machine learning techniques to enhance the interpretability of anomaly detection systems (Mohite and Ouarbya 2024).

2.4.3. Need for Advanced Algorithms

The demonstrated ineffectiveness of existing signature-based tools explained on [2.4.1](#) coupled with the development of sophisticated, AI-driven evasion techniques shown on [2.4.2](#) creates a significant security gap. As demonstrated by Suarez-Tangil, et al. (2014) and later Badhani and Muttoo (2018), traditional antivirus solutions are fundamentally ill-equipped to handle the dynamic and subtle nature of modern stegomalware, as they rely on detecting known patterns.

The evasion methods detailed, such as SGAN and EAST, are not merely hiding data; they are leveraging advanced machine learning concepts like Generative Adversarial Networks and adversarial attacks to actively fool detection systems (Ghamizi, et al. 2021). This represents a shift where attackers are using intelligent algorithms to create steganographic content that is statistically indistinguishable from legitimate media making the reactive signature-based defences no longer viable (Rafy 2024).

Dealing with these threats calls for a new type of detection algorithm that goes beyond simple pattern matching and uses adaptive, learning-based methods to uncover hidden data (Apau, et al. 2024). De La Croix, et al. (2024) described how such systems can combine statistical feature analysis, deep learning models, and anomaly detection tools to spot the small patterns left behind by AI-generated steganography. In addition, these models need to be resilient against adversarial attacks and able to adapt to new embedding methods, maintaining their effectiveness across different image changes and attack conditions (Alrusaini 2025). The next section examines how rule-based tools like YARA can form the foundation of hybrid detection systems that merge traditional pattern-matching with intelligent, learning-based methods for detecting anomalies.

2.5. YARA: Capabilities and Limitations

YARA is a powerful tool used for pattern matching and rule-based classification that has become a critical component in modern malware identification and mitigation (Vinay Patil 2025). YARA enables security professionals to create custom rules that detect specific patterns in files, memory, or network traffic, moving beyond the limitations of traditional hash-based indicators (NCCIC 2015). Understanding both the extensive capabilities and the inherent limitations of this signature-based approach is essential for using it effectively and for developing a comprehensive, layered defence against evolving cyber threats (Vinay Patil 2025). This section looks at the tool's roles and capabilities in relation to this study and further explore the limitations.

2.5.1. YARA's Role and Capabilities

YARA, often described as "The pattern matching Swiss knife for malware researchers", serves as a de-facto standard for malware identification and classification (VirusTotal 2019). Its primary function is to identify and classify malware by creating rules that search for specific patterns, which can be either textual or binary in nature (Lockett 2021). Unlike traditional cryptographic hashing that looks for exact file matches, YARA excels at identifying entire malware families by exploiting common code similarities (NCCIC 2015). As demonstrated by Coscia, et al. (2025), a YARA rule which is a created pattern, is composed of a meta section for descriptive information, a strings section where patterns are defined, and a condition section that contains a Boolean expression determining when the rule should trigger a match, this is further explained by Dittmann, et al. (2024) who discussed some of the key capabilities of YARA shown below:

Flexibility and Customization:

YARA is widely accepted for its flexible and customizable nature, which allows malware analysts to develop rules tailored to the requirements of a specific security domain (Naik, et al. 2020). Rules can be crafted with great precision using text strings, hexadecimal strings, and regular expressions to define patterns (Coscia, et al. 2025).

Platform Support and Integration:

It is an open-source, multi-platform tool that works on Windows, Linux, and Mac OS explained by Naik, et al. (2020) and further noted that YARA can be integrated as a library into other applications and programming languages, including popular antivirus engines like ClamAV, Python, and C/C++.

Versatility in Analysis:

The tool can be used for both static and dynamic malware analysis (Naik, et al. 2020). It is effective at detecting disguised malware, as rules can be built to identify specific obfuscation techniques like packing or anti-debugging capabilities (Lockett 2021).

Extensibility:

For conditions that are too complex for the default syntax, YARA's functionality can be extended with custom modules that provide additional metrics and functions not available in the default toolset (Dittmann, et al. 2024).

2.5.2. YARA's Limitations

Despite its widespread use and capabilities, YARA is not a complete solution and has several limitations, many of which are inherent to its nature as a signature-based detection tool (Lockett 2021).

Evasion by Attackers:

The effectiveness of YARA rules is dependent on their defined Indicator of Compromise (IoC) strings, but attackers can manipulate, encrypt, or replace these strings to evade detection (Naik, et al. 2020). Static rules that rely on fixed byte sequences are often ineffective against polymorphic or metamorphic malware, which is designed to constantly change its code structure to avoid signature matching (Vinay Patil 2025).

Rule Creation and Maintenance:

Writing high-quality, effective YARA rules is a tedious process that requires a highly specialized skill-set and expert knowledge of malware analysis. As new malware variants are constantly developed, the continuous maintenance, refinement, and updating of rule sets can be a complex and time-consuming task (Khalid, et al. 2020).

Balancing False Positives and Negatives:

Creating a good rule involves a difficult trade-off between sensitivity and precision. A rule that is too specific might miss new malware variants, while a rule that is too general can produce a high number of false positives by incorrectly flagging benign files (Vinay Patil 2025). The success of YARA depends on achieving a challenging balance of the right types and number of IoC strings (Naik, et al. 2020).

Detection of Novel Threats:

YARA is most effective in detecting already existing malware families and variants. It may fail to detect entirely new or unique malware strains for which no signatures have yet been created (Naik, et al. 2020).

Automated Rule Generation:

While tools that automatically generate YARA rules exist, their output often requires manual post-processing and optimization to be effective and may not perform as well as rules created manually by an expert analyst (Naik, et al. 2020).

2.5.3. The Need for a Hybrid Approach

Vinay Patil (2025) highlighted the necessity of adopting a hybrid strategy that combines YARA with other detection techniques to create a more resilient and effective defence due to the limitations YARA offers as a standalone signature-based tool. The limitations of YARA as a standalone, signature-based tool. They further describe how since static YARA rules can be bypassed, especially by evasive malware, they must be augmented with more dynamic and flexible methods.

One proposed hybrid solution is to enhance YARA rules by integrating them with fuzzy hashing (Naik, et al. 2020) which complements YARA by identifying structural similarities between files. This method is effective even when specific strings or byte sequences have been modified (Naik, et al. 2020) and this combined search can increase the accuracy of detection and the overall confidence in the analysis results as Naik, et al. (2020) described and further research by Lockett (2021) has also suggested building fuzzy hashing functionality directly into YARA rules which can help in strengthening the rules.

Furthermore, the weakness of signature-based detection against new and evasive threats highlights the need to incorporate behaviour-based analysis (Lockett 2021). A hybrid strategy that combines YARA's static pattern matching with dynamic analysis which focuses on runtime actions such as suspicious API calls, system modifications, and network activity is far more effective at spotting complex risks (Vinay Patil 2025). This study explores the integration of machine learning with YARA to automatically generate more adaptive and intelligent rules, helping to overcome the difficulties and time constraints of manual rule creation (Vinay Patil 2025).

2.6. The Role of VirusTotal in Testing

VirusTotal is a free online tool that one of the most popular malware scanning platforms (Leka, et al. 2022) which plays a crucial role in the testing and analysis of potentially malicious files and URLs (Masri and Aldwairi 2017). Its primary function is to analyse submissions using a vast collection of different antivirus engines, website scanners,

and domain blacklists to detect various types of malwares, including viruses, worms, and trojans (Masri and Aldwairi 2017).

VirusTotal contains the YARA framework which highlights the organization's key role, ongoing contribution to the malware research community and its central role in the malware research community (Dittmann, et al. 2024). In experimental and research settings, VirusTotal is frequently used as a definitive source to validate whether samples are malicious and to verify their correct classifications which can be achieved uploading script of a URL or binary and retrieve the "detection ratio" (number of engines that label it malicious) (Lockett 2021). Its extensive database of known malware hashes and domains makes it invaluable for validation as it allows researchers frequently cross-reference their samples against VirusTotal to ensure correct labelling (Masri and Aldwairi 2017).

Beyond manual uploads, and apart from its primary public interface, PC uploaders, browser extensions, and an Application Programming Interface (API) are all viable avenues for the transmission of data to VirusTotal (Vasani, et al. 2023). A system can be designed to automatically submit URLs to VirusTotal and parse the detection ratio to classify them as malicious or benign (Masri and Aldwairi 2017). Its extensive and reliable threat database utilizes more than 70 anti-malware analysers and URL/domain blacklisting services (Vasani, et al. 2023). However, while it is indispensable for testing, its performance is not foolproof as Leka, et al. (2022) explained there is no clear understanding of how it delivers its results and since the platform operates basically as a Blackbox thus the produced results may be questionable. In one study evaluating its effectiveness in detecting malicious advertisements by Masri and Aldwairi (2017) VirusTotal achieved a testing accuracy of 56%, which, while significant, was outperformed by other tools such as URL void with 73% that incorporated a wider range of domain blacklists in addition to scanning engines and highlight that while VirusTotal is a powerful and widely used testing platform, its effectiveness can be enhanced when used as part of a broader, multi-tool testing strategy.

The selection of VirusTotal as the primary validation tool for this study is justified by it been a free and accessible online service, it presents a simple yet powerful solution for large-scale analysis, eliminating the complexities and costs associated with acquiring and maintaining dozens of individual antivirus products. This study will leverage its Application Programming Interface (API) to systematically upload AI-generated steganographic samples containing hidden payloads. The resulting "detection ratio" metric indicating how many of the 70+ integrated, industry-leading antivirus engines flag a sample as malicious will serve as the core measure of evasion success. A low detection ratio will provide powerful, empirical evidence of the AI steganography's ability to bypass the current threat detection landscape. Therefore, Virus Total's comprehensive and standardized approach provides a credible, large-scale benchmark that is essential for strengthening the experimental validation component of this research and solidifying its conclusions on vulnerabilities within modern antivirus systems.

2.7. Emerging Solutions and Approaches

So far on this chapter, it has been noted how using steganography during an attack is relatively easy but protecting against it is much more complicated as threat actors are getting more innovative and more creative (Kaspersky 2023). The core challenge lies in the fact that steganography's primary goal is to hide the very existence of a secret message, making it difficult to detect but they do leave behind detectable traces in the cover medium (Provos and Honeyman 2003).

As noted before, traditional antivirus systems use signature based detection, which relies on a vendor-generated database (Raheel Abbas and Hussain 2025). But as Raheel Abbas and Hussain (2025) further demonstrated, that this method is only effective against known threats and with the rapid evolution of malware, particularly zero-day attacks, signature-based detection often falls short in identifying new or mutated threat. This steganography vulnerability has necessitated the development of more sophisticated and multi-layered defence frameworks (Prathyusha, et al. 2025) and some emerging solutions are moving beyond simple signature matching to embrace more intelligent methods of analysis, focusing on areas such as targeted pattern-matching of custom YARA rules, statistical anomaly detection of machine learning, and synergistic power of hybrid models.

2.7.1. Custom YARA Rules

YARA is a tool aimed at (but not limited to) helping malware researchers to identify and classify malware samples (Ningappa 2020). It functions by creating custom rules that match specific, predefined patterns within a file's binary or text content (Vinay Patil 2025). This signature-based detection approach is a fundamental technique used by antivirus and malware analysis software (Lockett 2021). The value of custom rules lies in their ability to be tailored to the unique characteristics of specific threats, such as the small artifacts left by steganography (Khalid, et al. 2020). While YARA is a widely accepted and flexible tool for identifying known patterns, its effectiveness is limited against polymorphic malware, which are designed to constantly change their signature to evade detection (Vinay Patil 2025). Despite this limitation, custom YARA rules remain an essential and efficient tool for threat hunting and providing a rapid method for identifying known malicious indicators (NCCIC 2015).

The study will follow process described by Naik, et al. (2020) who explained how YARA at its core is designed to spot malware by matching its characteristics, or "signatures," against a database of known threats. They further explain how it works by scanning files and processes for these tell-tale signs and built with three main parts which are the meta, strings, and condition as noted below in an example of a simple YARA Rule syntax they employed:

```

rule RuleName {

    meta:
        description = "description of rule"
        author = "name"
        date = "dd/mm/yyyy"
        reference = "url"

    strings:
        $text_string1 = "text1 you wish to find in malware"
        $text_string2 = "text2 you wish to find in malware"
        $hex_string1 =
        $hex_string2 =
        $reg_exp_string1 = /regular expression1 you wish to find in malware/
        $reg_exp_string2 = /regular expression2 you wish to find in malware/

    condition:
        $text_string1 or $text_string2 or
        $hex_string1 or $hex_string2 or
        $reg_exp_string1 or $reg_exp_string2
}

```

Regeciova (2022) explained the different sections involved in detail from the strings section is where patterns been hunted are defined which can further fall into three categories:

Text Strings:

Simple, readable text. You can add modifiers to make the search case-insensitive (no case) or specify the character type (ascii, wide).

Hexadecimal Strings:

Used for matching specific sequences of raw bytes, which is great for finding non-textual data. This format is flexible, allowing for wildcards and other advanced patterns.

Regular Expressions:

A very powerful option, available since version 2.0, that allows you to define complex and variable patterns.

The final piece they explained is the condition section. This is the rule's brain, using a logical statement e.g. if 2 of the 3 strings are found, then it's a match to decide whether to officially flag the file as malware. It's this condition that ultimately determines if the rule triggers.

2.7.2. Machine Learning-Based Anomaly Detection

To address the limitations of static signatures discussed earlier, emerging solutions are increasingly turning to machine learning (ML) (Adhav, et al. 2024). Machine learning gives computers the ability to learn from data without being explicitly programmed, allowing them to distinguish complex patterns that traditional methods miss. In particular, unsupervised learning systems can analyse unlabelled data to discover hidden patterns and classify outliers on their own (Naeem, et al. 2023).

ML-based anomaly detection identifies threats by detecting behaviours and observations that deviate significantly from a baseline of normal activity, rather than looking for a known signature (Zehra, et al. 2023). This approach is particularly effective against unknown or “zero-day” threats because it does not rely on prior knowledge of a specific attack (Guo 2023). This makes ML-based anomaly detection a powerful method for flagging suspicious files based on statistical irregularities and unusual behaviours that might indicate concealed malware (Ndibe 2025). As a result, anomaly detection enhances rule-based tools like YARA by adding adaptive intelligence that can identify new and previously unseen attack patterns (Quffa and Abu-Naser).

2.7.3. Hybrid Detection Models

Recognizing the limitations of both signature-based and purely behavioural systems, researchers are increasingly turning to hybrid detection models as the most effective emerging solution (Vinay Patil 2025). A hybrid approach combines the speed and efficiency of signature-based tools like YARA for known threats with the adaptability of machine learning for unknown and evasive malware (Adhav, et al. 2024). This layered strategy creates a more comprehensive defence, using different techniques to cover a wider range of attack vectors (Vinay Patil 2025). Such an integrated framework combining static analysis with ML-based methods, is designed to improve overall detection accuracy while simultaneously reducing the rate of false positives by leveraging the complementary strengths of each method, a hybrid model can provide a more resilient and effective defence against the evolving threat landscape (Naik, et al. 2020).

2.8. Summary

This literature review examines the existing body of knowledge on steganography and malware detection to provide the foundational context for the rest of the study. The review shows the evolution of steganography, highlighting its progression from historical practices to modern, AI-driven digital concealment techniques that present significant detection challenges. A key takeaway is that most antivirus tools still depend on old-style signature detection, which often can't catch malware once it's hidden inside multimedia files.

This critical detection gap is substantiated by studies by the likes of Gao, et al. (2022) and Almomani, et al. (2022a) demonstrating that even known malware, such as ransomware, becomes invisible to dozens of antivirus engines after being embedded using steganographic techniques. The review also assesses the capabilities and inherent limitations of YARA, a key tool for pattern-based malware detection. While YARA offers flexibility in developing custom rules, its effectiveness as a signature-based tool is inconsistent against the evasive and evolving nature of steganographic threats.

Finally, the review explores emerging solutions that aim to overcome these challenges. There is a clear trend towards more adaptive systems, with a strong emphasis on machine learning-based anomaly detection as a promising approach for identifying concealed threats through statistical irregularities rather than known signatures. The findings confirm a distinct gap in current detection methodologies and strongly support the rationale for this study's proposed hybrid model, which integrates enhanced YARA rules with machine learning to counter the threat of AI-generated steganographic malware.

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